

VƯƠNG VIỆT HÙNG

PERSONAL DETAILS

Full name:	Vương Việt Hùng	Nationality:	Vietnamese
Phone:	+84-583-885-391	Languages:	Vietnamese (Native), English (IELTS 7.5)
Email:	vuongviethung156@gmail.com	ORCID:	 0009-0006-8041-7608
Residence:	Hanoi, Vietnam	Google Scholar:	Vương Việt Hùng

EDUCATIONAL QUALIFICATIONS

Soongsil University	Seoul, South Korea
Master's Degree in Electronic Engineering	March 2024 – March 2026
– GPA: 4.24/4.50	
– Thesis: Scene Flow-Guided Online Continual LiDAR Panoptic Tracking	
– Thesis length: 10,387 words	

Hanoi University of Science and Technology	Hanoi, Vietnam
Bachelor's Degree in Control Engineering and Automation	August 2019 – September 2023
– GPA: 3.34/4.00	
– Thesis: Stereo Matching and 3D Reconstruction	
– Thesis length:	

EMPLOYMENT HISTORY

VinDynamics	Hanoi, Vietnam
SLAM & Navigation Engineer	March 2026 – current
Developed and deployed SLAM and height mapping systems for humanoid robotic platforms, with a focus on sensor integration, system evaluation, and optimization.	

ANDA Lab, Soongsil University	Seoul, South Korea
Research Assistant	March 2024 – March 2026
Conducted research in LiDAR panoptic segmentation, scene completion, domain adaptation, and continual learning; contributed to peer-reviewed publications, researcher mentoring, and technical seminars.	

Rikai Mind	Hanoi, Vietnam
AI Engineer Intern	August 2022 – January 2023
Developed and deployed a computer vision system for object volume estimation from monocular images and camera-object distance.	

Mandevices Laboratory, Hanoi University of Science and Technology	Hanoi, Vietnam
Research Assistant	August 2020 – September 2023
Conducted research in deep-learning-based bearing fault diagnosis and contributed to data collection, publications, and researcher mentoring.	

RESEARCH OUTPUTS

Journal Articles

- Vuong, H. V., & Yoo, M. (2026).** Domain adaptive LiDAR panoptic segmentation via pseudo-supervised instance learning. *IEEE Transactions on Intelligent Transportation Systems*.
<https://doi.org/10.1109/TITS.2026.3695516>
Journal Impact Factor: 9.1
 - Proposed PILLA, a domain adaptation framework that generates instance-level pseudo-labels via unsupervised clustering and applies a noise-aware loss for robust pseudo-supervision.
 - Achieved 35.97% PQ on nuScenes in cross-dataset adaptation from SemanticKITTI, outperforming AdaptLPS by 13.14 percentage points.

2. Ha-Phan, N.-Q., **Vuong, H. V.**, & Yoo, M. (2025). Instance embedding as queries for DETR-based LiDAR panoptic segmentation. *IEEE Transactions on Intelligent Vehicles*, 10(10), 4629–4641.

<https://doi.org/10.1109/TIV.2024.3488035>

Ha-Phan, N.-Q. and Vuong, H. V. contributed equally to this work.

Journal Impact Factor: 14.3

- Proposed a DETR-based architecture that uses instance embeddings as object queries for LiDAR panoptic segmentation.
- Achieved 63.40% PQ on SemanticKITTI and 77.08% PQ on nuScenes, outperforming previous methods by 1.20 and 1.94 percentage points, respectively.

3. Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2025). FuPaSCo: Long-range and local context fusion for 3D panoptic scene completion. *Image and Vision Computing*, 163, 105776.

<https://doi.org/10.1016/j.imavis.2025.105776>

Journal Impact Factor: 4.2

- Developed FuPaSCo, a dual-branch CNN–Transformer framework that combines local and long-range features through feature fusion and alignment for LiDAR panoptic scene completion.
- Achieved 26.10% PQ[†] on SemanticKITTI, outperforming PaSCo by 2.49 percentage points.

4. Nguyen, T. H., **Vuong, V. H.**, Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). Generalized simulation-based domain adaptation approach for intelligent bearing fault diagnosis. *Arabian Journal for Science and Engineering*, 49(12), 16941–16957. <https://doi.org/10.1007/s13369-024-09282-1>

Journal Impact Factor: 3.1

- Developed a simulation-based domain adaptation approach that generates synthetic vibration signals from bearing geometry to address data scarcity in intelligent bearing fault diagnosis.
- Achieved 74–98% fault classification accuracy across different bearing types using a CNN trained on synthesized vibration signals.

Published Conference Proceedings

1. Nguyen, T. H., Le, H. A., **Vuong, V. H.**, Tran, L. D. A., Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). ConvMamba: A data-efficient neural network for bearing fault diagnosis. In *2024 7th International Conference on Green Technology and Sustainable Development (GTSD)* (pp. 165–172). IEEE.

<https://doi.org/10.1109/GTSD62346.2024.10674880>

- Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba for bearing fault diagnosis.
- Improved classification accuracy by 27 percentage points over the predecessor Hybrid-CNN-MLP model on the CWRU bearing fault dataset.

Manuscripts Under Review

1. Pham, T. P., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). SM3-PSC: Sparse multi-modal Mamba for 3D panoptic scene completion with multi-scale feature refinement. *Manuscript submitted for publication*.

- Developed SM3-PSC, a sparse multi-modal framework that fuses RGB semantics with LiDAR voxels through deformable cross-attention and uses Mamba for efficient long-range context modeling.
- Achieved 28.20% PQ[†] on SemanticKITTI, outperforming FuPaSCo by 2.10 percentage points while running the 3D pipeline 2.8× faster.

2. Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). LW-PSC: Lightweight design for LiDAR panoptic scene completion. *Manuscript submitted for publication*.

- Developed LW-PSC, a lightweight BEV-based framework that combines Cartesian and Polar representations with center-focused feature refinement for real-time LiDAR panoptic scene completion.

- Achieved 13.59% PQ at 11.3 FPS on SemanticKITTI, outperforming PaSCo by 1.47 percentage points while reducing the model size from 40.65M to 29.16M parameters.